OPTION C • SERVERLESS CI/CD BOT

CI/CD AUTOMATION BOT & ZERO-TOUCH FLOW

Codebase-Introspective Hybrid Quality Engine • Self-Healing E2E • Automated Jira Delivery

Platform Codebase: 0 Lines Added (Strict Read-Only) | Idle Cost: $0/month | Wakeup: Event-Driven

SECTION 1 • WHAT BOT DO WE USE?

RUNNER — SERVERLESS RUNNER BOT

GitHub Actions / GitLab Runner

- Completely Serverless in Cloud
- Wakes up in < 5s on events
- Isolated container environment
- Powers down after run (0% CPU)
- Zero local PC / laptop dependency

TRIGGER — SMART DIFF TRIGGER BOT

Git Webhooks & S3 Watcher

- Wakes on Code Merge or S3 Run
- Executes git diff HEAD~1
- Skips docs/markdown (exits in 3s)
- Identifies impacted tools/routes
- Guarantees targeted test runs

JIRA BOT — JIRA CLOUD API BOT

Jira REST API v3 Integration

- Automated artifact uploads
- Executive PDF + Excel reports
- Formats native pills [^filename.ext]
- Updates QA tickets automatically
- 100% visible to entire team

SECTION 2 • THE 7-STAGE EXECUTION LIFECYCLE

1	2	3	4	5	6	7
TRIGGER	WAKE UP	DIFF FILTER	SENSORS	VALIDATE	REPORTS	JIRA POST
Push / PR or S3 Event	Runner Boots in < 5s	git diff HEAD~1	Read AST & Next.js Routes	Whitebox + Self-Heal E2E	PDF Review + Excel Tracker	Upload Pills Power Down 0%

✦ 100% Autonomous Execution: 25-35 Seconds Total Duration • $0 Idle Cost • Zero Human Intervention

SECTION 3 • ZERO-TOUCH PLATFORM CODEBASE ARCHITECTURE

PLATFORM CODEBASE (READ-ONLY)

F:\ai-testable-platform

[X] 0 Lines of Code Added

[X] 0 Test Dependencies Installed

[X] 0 Commits or Branches Created

[!] Mounted 100% Read-Only

[*] Contains Ground Truth Registries

[*] Contains Live Next.js App Routes

READ-ONLY SENSORS (1-WAY FLOW)

- **Sensor 1: Python AST Tool Reader**
  Reads technique_tool.py directly.
  Auto-discovers newly registered tools.
- **Sensor 2: Next.js Route Scanner**
  Scans frontend/web-console/src/app/
  Maps live page.tsx URLs automatically.
- **Sensor 3: Git-Diff Impact Filter**
  Runs git diff HEAD~1 against commit.
  Targets only affected test suites.

AUTOMATION BOT ENGINE

F:\Tool Validation Automation

- Isolated Framework Repository
- Self-Healing Action Resolver
- Playwright Browser Driver
- Multi-Sheet Excel Engine
- Executive 2-Page PDF Builder
- Jira REST API Cloud Poster

SECTION 4 • SMART GIT-DIFF TRIGGER DECISION TREE

EVENT FIRED

Push / PR Merge / S3

git diff HEAD~1

Inspect Changed Files

NO-OP	Docs / Readme Only (*.md, docs/*, .gitignore)	Immediate Power-Down (3s) → Zero tests run · $0 cost
WHITEBOX	Static Tools Modified (shared/scoring/*, tools/*)	Re-Sync Tool AST → Execute Whitebox & Taxonomy Parity Validation
E2E UI	Frontend Modified (frontend/web-console/*)	Scan Next.js Routes → Execute Self-Healing Playwright E2E Tests
FULL RUN	Backend Orchestrator (backend/wb-orchestrator/*)	Verify Celery Scheduling → Full End-to-End Regression & Container Validation

SECTION 5 • SELF-HEALING DYNAMIC UI RESOLUTION ENGINE

RENAME — BUTTON RENAMED

e.g. 'Login' → 'Continue'

Mechanism:

Semantic Intent Engine

Resolves by universal intent (AUTH, TRIGGER, EXPORT). Matches synonyms automatically.

POSITION — LOCATION SHIFT

Moved to Header or Sidebar

Mechanism:

Accessibility DOM Tree

Zero pixel dependency (no X/Y). Auto-scrolls viewport to bring element into clickable view.

DROPDOWN — 3-DOTS MENU (...)

Row Action Dropdown (...)

Mechanism:

Auto-Kebab Unfurler

Detects hidden button, clicks parent row action trigger, unfurls menu, and clicks item.

ICON ONLY — ICON-ONLY SUBMIT

Arrow → with Zero Text

Mechanism:

Form Hierarchy Anchor

Detects active form container and targets form button [type='submit'] by functional role.

SECTION 6 • HOW THE ENTIRE TEAM COLLABORATES

DEV TEAM — DEVELOPERS

Benefit: 0 Script Installation

- Push code or open PRs normally
- Bot validates static tools automatically
- Receives instant PR status checks
- Clear root-cause attribution on failure

QA TEAM — QA ENGINEERS

Benefit: 0 Code Scripting

- Add E2E tests in Excel (testcases/admin.xlsx)
- Zero python/playwright coding required
- Inspect container defect classification
- Verify taxonomy parity gates effortlessly

LEADERSHIP — LEADERSHIP / MANAGERS

Benefit: 1-Click Executive Delivery

- Jira tickets updated with 2-page review PDFs
- Excel matrices attached with native pills
- 30-second executive BLUF summary
- Saravanan / Rathiba / Benny QA standard

EXECUTIVE QA ARCHITECTURE SIGN-OFF

SARAVANAN SELVARAJ • RATHIBA • BENNY QA STANDARD COMPLIANCE

- ✓ 100% Non-Destructive Safety: Platform repository is strictly Read-Only. 0 code changes, 0 dependencies, 0 commits.
- ✓ Zero Maintenance Overhead: Self-healing semantic actions & AST introspection adapt to UI & tool changes automatically.
- ✓ True Serverless Architecture: GitHub Actions runs on-demand for ~30 seconds with $0 idle cost and 0% CPU consumption.
- ✓ Automated Jira Cloud Delivery: Native pills [^filename.ext] with clickable reports, executive PDFs, and Excel trackers.